

Statistical Analysis of IPL Match Data

Descriptive Statistics, the Binomial Model, and Regression

Rajagiri College of Social Sciences (Autonomous)

A case study on IPL match data from 2008–2018

Group members

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- Dataset and problem statement
- Descriptive statistics and visualisation
- Binomial probability modelling
- Range-based probabilities
- Regression: scoring across the decade
- Conclusions and limitations

Dataset and Problem Statement

Source: supplied IPL match archive, 2008–2018.

- 1169 total matches in archive
- Window 2008–2018 (seasons 1–11): 696 matches
- 10 removed (no result) $\Rightarrow$ **686 decided matches**
- 10 of 19 variables retained

Objectives

- 1 Summarise matches (central tendency, dispersion)
- 2 Model the win process with the Binomial distribution
- 3 Compute range-based probabilities
- 4 Test for a scoring trend with linear regression

Selected variables

Variable	Role
season	time index
toss_decision	bat / field
winner	outcome
result	runs / wickets / tie
result_margin	margin
target_runs	first-innings score

Descriptive Statistics

Target runs ($n = 686$)

Mean 161, median 163, mode 166

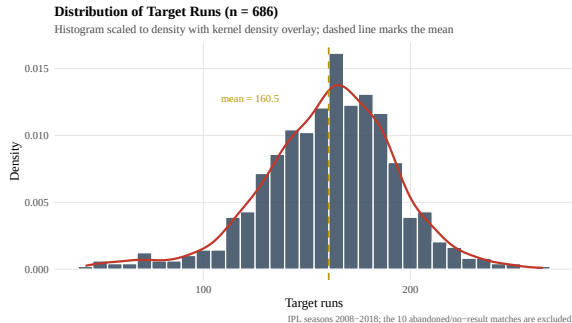
SD 32.4 CV 0.20

Victory margin ($n = 685$)

Mean 17, median 8

SD 22.0 CV 1.29

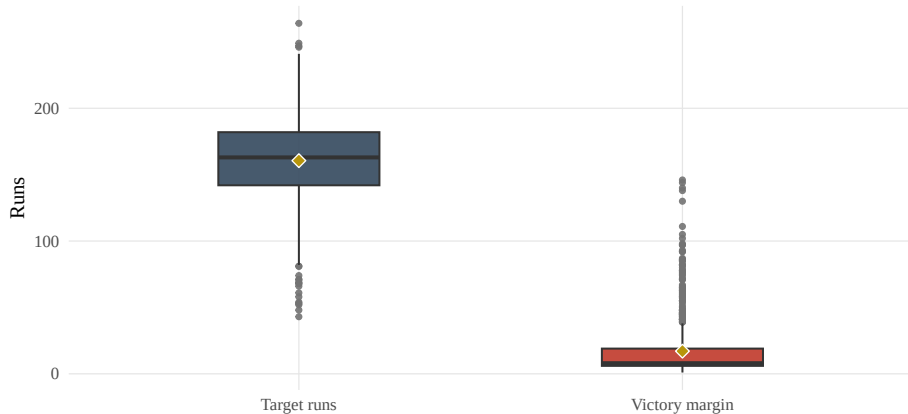
Margins are strongly right-skewed (skew 2.8):
most games are close, blow-outs are rare — yet
the maximum reaches 146 runs.



Dispersion: Box-and-Whisker Comparison

Spread and Skew: Victory Margins vs Target Runs

Boxes span Q1–Q3; median line and gold mean diamond shown.
The margin box is compressed, with far-out outliers.



Margins available for 685 decided matches (the single tied match has no margin).

The Binomial Model: First-Innings Advantage

Each match is a Bernoulli trial with p = win probability; counts follow $X \sim \text{Binomial}(n, p)$.

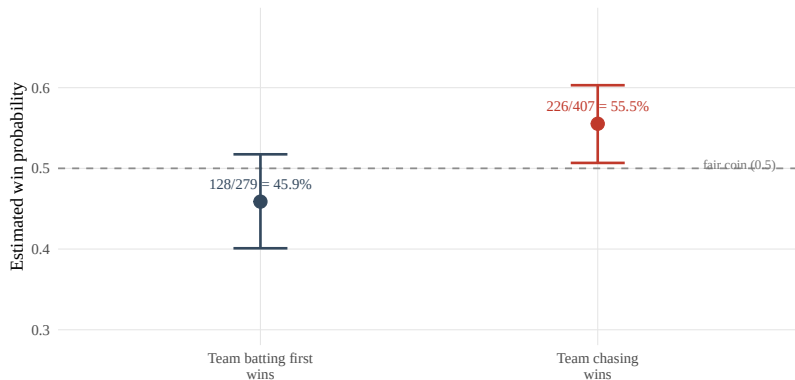
Strategy	wins / n	$\hat{p}$	p -value ($H_0: p = 0.5$)
Bat first	128 / 279	0.459	0.188
Chase	226 / 407	0.555	0.029

Chasing estimate is **significantly** above 0.5 ($p = 0.029$):
a real, modest advantage for the fielding side this decade.

Binomial Estimates with 95% Confidence Intervals

First-Innings Advantage: Binomial Proportion Estimates

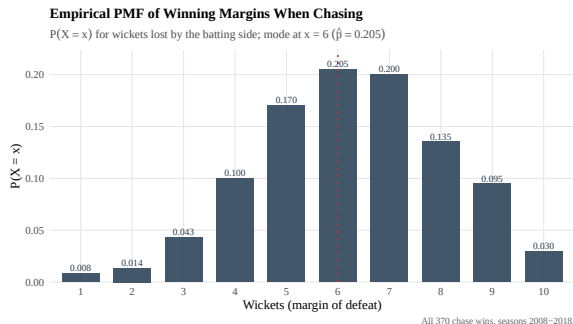
Win proportions with 95% confidence intervals; chasing clears the fair-coin benchmark, batting first does not.



Exact binomial tests: chasing $p = 0.029$ (significant at $\alpha = 0.05$); batting first $p = 0.188$.

The chasing interval lies entirely above the fair-coin line (0.5); the batting-first interval straddles it.

Probability Mass Function of Wicket Margins



When a chase succeeds, the margin is *wickets in hand*, a discrete variable $X \in \{1, \dots, 10\}$.

- Mode at $X = 6$: $P(X = 6) = 0.205$
- Unimodal, concentrated between 3–8
- Rare to win by 1 or 10 wickets

Range-Based Probabilities

Where closed-form integrals are not available, probabilities are estimated by relative frequency:

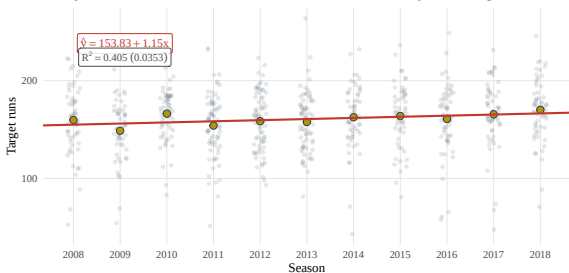
Event	Estimate
$P(140 \leq T \leq 160)$	0.233
$P(161 \leq T \leq 190)$	0.389
$P(T > 200)$	0.087
$P(T < 120)$	0.093
$P(\text{margin} > 50)$	0.074

Most targets fall in the 140–190 band; scores above 200 are uncommon.

Regression: Scoring Across the Decade

Target Runs Drift Upward Across the Decade

Faded points: individual matches. Gold dots: seasonal means. Red line: least-squares fit through seasonal means.



OLS on 11 seasonal means: slope = 1.150 runs per season (SE 0.465).

Least-squares fit on 11 seasonal means:

$$\hat{y} = 153.9 + 1.15x$$

- Slope +1.15 runs/season
- $R^2 = 0.405$
- $p = 0.035$ (significant at 5%)

First-innings targets drifted *upward* across 2008–2018.

Conclusions

- Targets concentrate near 160–165 and rise ≈ 1.2 runs/season
- Margins are heavily right-skewed — T20 is a game of close finishes
- **Chasing wins more often than not** ($\hat{p} = 0.555$, $p = 0.029$)
- Toss alone is still a coin flip (no inherent luck)
- Team dominance is real but measured (CSK 90/146, $p = 0.006$)

Limitations & Reflection

- Relative-frequency estimates assume a representative sample; conclusions describe the 2008–2018 era, not cricket in general
- Ten variables collapse rich covariates (venue, opponents, pitch)
- A venue/opponent-aware model would refine the estimates
- Techniques map directly to the curriculum: PMF, Binomial, range probabilities, regression

Thank you.